CSE4204-8D-T03 Team Information

Team Name	CSE4204-8D-T03
Section	8D
Project Title	Smart Farmer Assistant: AI-Based Agriculture Support System
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GitHub Repository	**https://github.com/sakib-foysal/CSE4204-8D-T03-Smart-Farmer-Assistant**

Software Requirement Specification (SRS)

Smart Farmer Assistant: AI-Based Agriculture Support System

1. Introduction

i. Project Overview

Smart Farmer Assistant is an AI-powered agriculture support system designed to assist farmers in making better agricultural decisions through intelligent technologies. Agriculture is one of the major sectors of Bangladesh, and many farmers face challenges due to crop diseases, improper fertilizer usage, sudden weather changes, flood damage, and lack of access to agricultural experts.

This system aims to solve those problems by integrating Artificial Intelligence, Machine Learning, image processing, weather forecasting APIs, and chatbot technologies into a single platform.

The proposed system allows farmers to upload crop images and receive AI-generated disease predictions with treatment recommendations. Users can also interact with an AI chatbot for farming guidance, view weather forecasts and flood alerts, receive fertilizer suggestions, and track real-time market prices.

The platform is designed as a mobile-friendly web application so that farmers can access agricultural services easily using smartphones or web browsers.

ii. Objectives

Primary Objectives

- Build an AI-powered smart farming assistant
- Detect crop diseases using AI image analysis
- Provide intelligent fertilizer recommendations
- Integrate AI chatbot support using Gemini AI
- Provide weather forecasts and flood alerts
- Track real-time market prices
- Support Bangla language interaction
- Develop a responsive platform

Secondary Objectives

- Reduce agricultural losses
- Improve crop productivity
- Increase farmer awareness
- Provide affordable AI-based support
- Promote smart agriculture in Bangladesh

iii. Scope

a. Current Scope

User Management

- Registration
- Login
- Profile management
- Authentication system

Disease Detection

- Upload crop images
- AI image processing
- Disease identification
- Treatment suggestions

AI Chatbot

- Farming assistance
- Question answering
- Smart recommendations

Weather Service

- Weather forecast
- Flood notifications

Market Module

- Live crop prices

Fertilizer Module

- Fertilizer recommendations

Language Support

- Bangla interaction

b. Future Scope

- IoT sensor integration
- Smart irrigation system
- Drone crop monitoring
- AI crop yield prediction
- Farmer marketplace
- Offline AI prediction support

2. Functional Requirements

i. User Registration

Users can create accounts using:

- Name
- Email
- Password
- Mobile number

System Process:

- Validate information
- Store user data
- Create account

ii. User Login

Users can login using:

- Email
- Password

System Process:

- Verify credentials
- Authenticate users
- Grant dashboard access

iii. Profile Management

Users can:

- Update profile information
- Change passwords
- Manage account settings
- Disease Detection Module

iv. Crop Image Upload

Users can:

- Upload crop images

- Capture images from devices

v. Disease Detection

System performs:

- Image processing
- TensorFlow prediction
- Disease classification

vi. Recommendation Generation

System displays:

- Disease name
- Confidence score
- Treatment suggestions
- AI Chatbot Module

vii. Chatbot Interaction

Users can:

- Ask farming questions
- Receive AI-generated responses

System Process:

- Send request to Gemini AI
- Generate intelligent response
- Weather Module

viii. Weather Information

System shows:

- Temperature
- Rain probability
- Humidity
- Weather forecast

ix. Flood Alert

System warns users regarding:

- Flood risks
- Extreme weather conditions
- Fertilizer Module

x. Fertilizer Recommendation

System analyzes:

- Crop information
- Disease information

System provides:

- Fertilizer suggestions
- Market Module

xi. Market Price Tracking

System displays:

- Crop prices
- Market trends

3. Non-Functional Requirements

Non-functional requirements define quality attributes of the system.

i. Performance Requirements

- API response time below 3 seconds
- Prediction generation below 5 seconds
- Fast image processing

ii. Security Requirements

- JWT Authentication
- Password encryption

- Secure API communication
- User authorization

iii. Usability Requirements

- Simple user interface
- Easy navigation
- Bangla language support
- User-friendly design

iv. Reliability Requirements

- Operate continuously
- Minimize failures
- Ensure stable service

v. Scalability Requirements

- Additional users
- Future AI modules
- Cloud expansion

vi. Compatibility Requirements

- Android devices
- Desktop browsers
- Tablets
- Mobile browsers

4. User Roles

The proposed system contains multiple user roles.

i. Farmer/User

Responsibilities:

- Register account
- Login

- Upload crop images
- Ask AI questions
- View weather information
- Receive recommendations

ii. Admin

Responsibilities:

- Manage users
- Monitor system activities
- Update market information
- Maintain system

iii. AI System

Responsibilities:

- Disease prediction
- Recommendation generation
- Chatbot responses
- Data processing

5. Use Cases

Use Case 1: User Registration

User Actions (Farmer):

- Open registration page
- Enter personal information
- Submit form

System Responses:

- Validate information
- Create account
- Redirect user

Use Case 2: User Login

User Actions (Farmer):

- Enter login credentials
- Click login button

System Responses:

- Verify credentials
- Authenticate user
- Open dashboard

Use Case 3: Disease Detection

User Actions (Farmer):

- Upload crop image
- Submit image

System Responses:

- Analyze image
- Detect disease
- Generate recommendation
- Display result

Use Case 4: AI Chat Assistant

User Actions (Farmer):

- Ask question
- Submit message

System Responses:

- Process question
- Generate response
- Display suggestion

Use Case 5: Weather Information

User Actions (Farmer):

- Open weather page

System Responses:

- Retrieve weather data
- Display forecast

6. Diagrams

i. Use Case Diagram

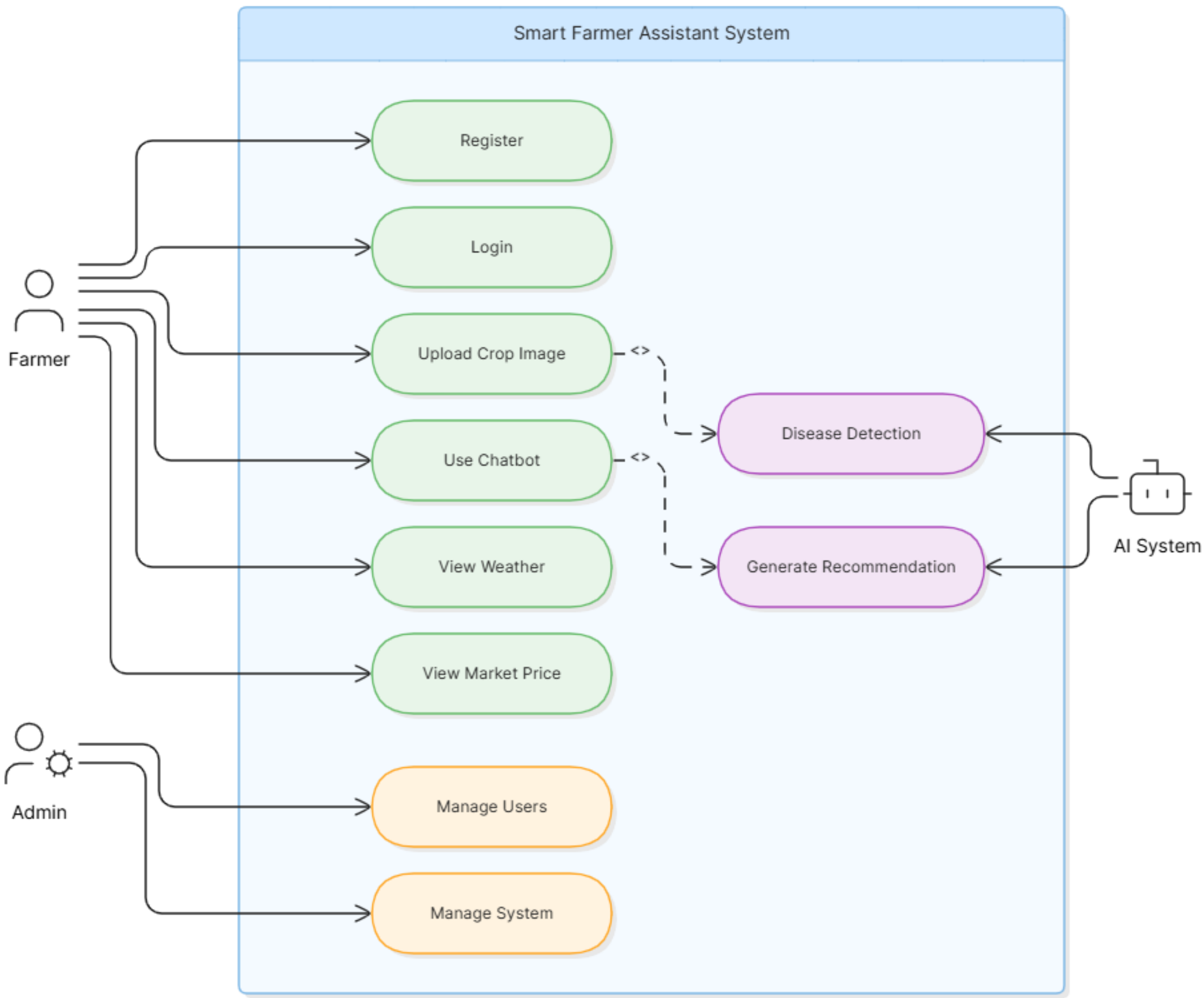

Fig-01: Use Case Diagram

ii. System Architecture Diagram

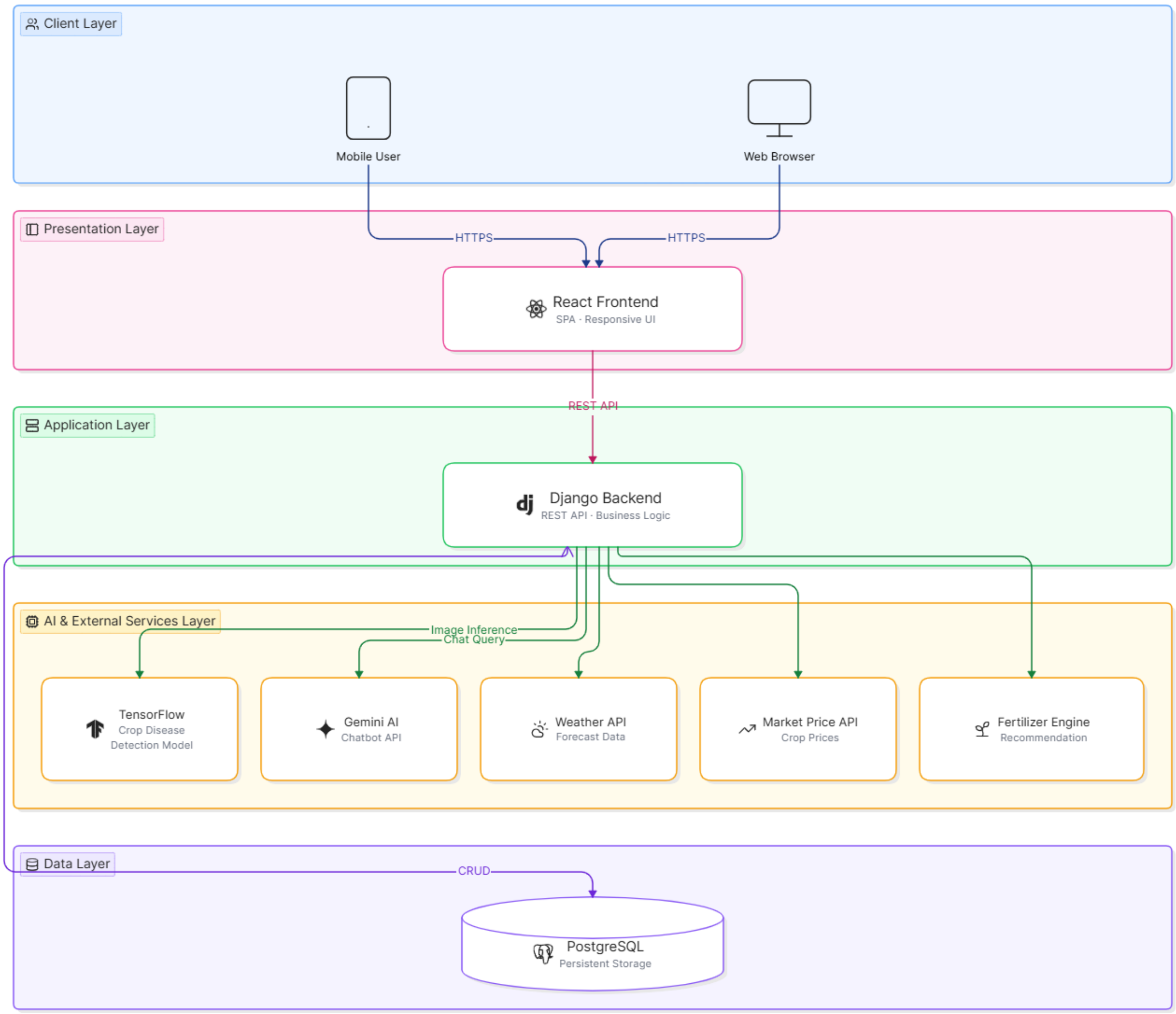

Fig-02: System Architecture Diagram

iii. Workflow Diagram

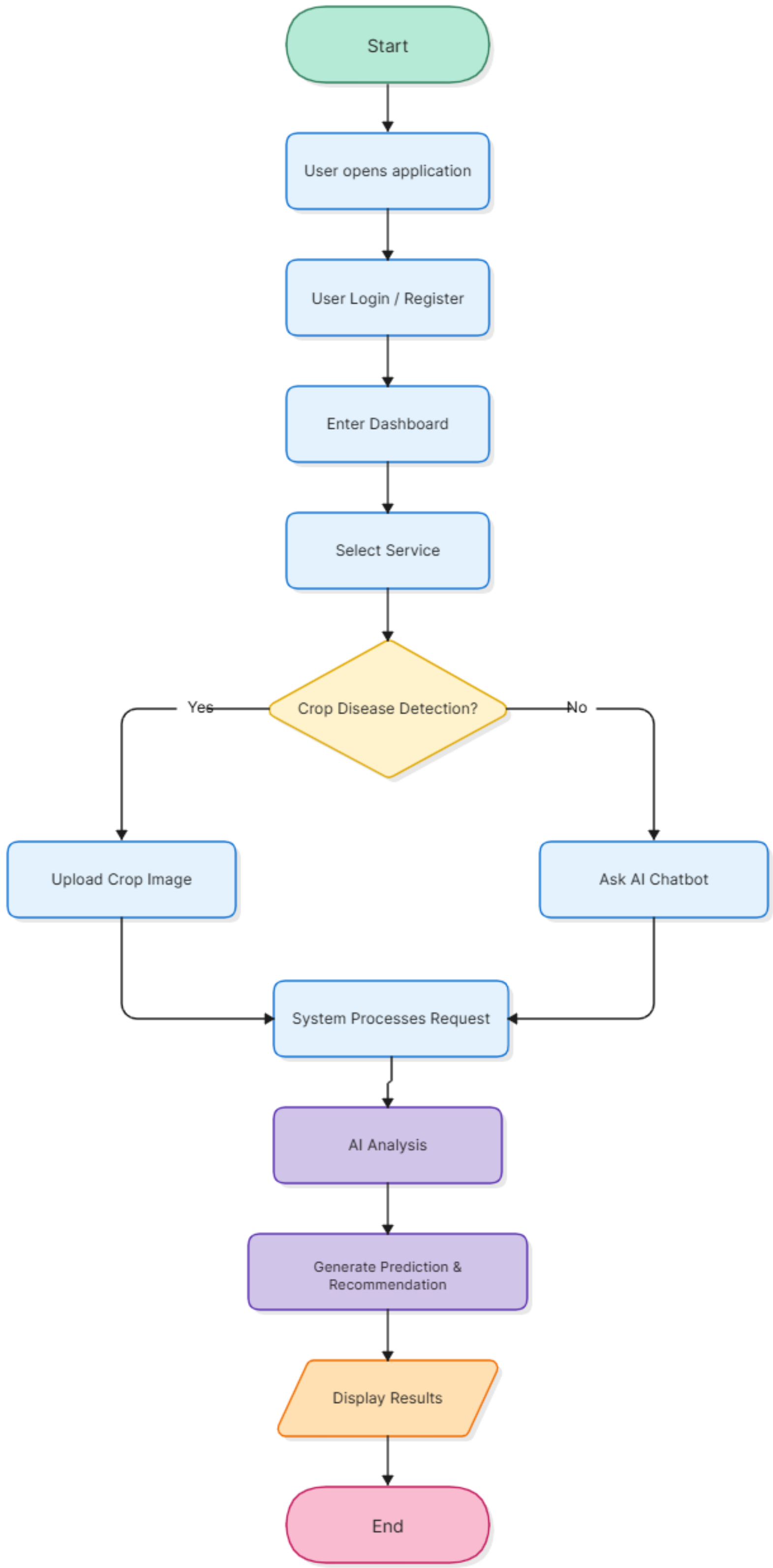

Fig-03: Workflow Diagram

iv. ER Diagram

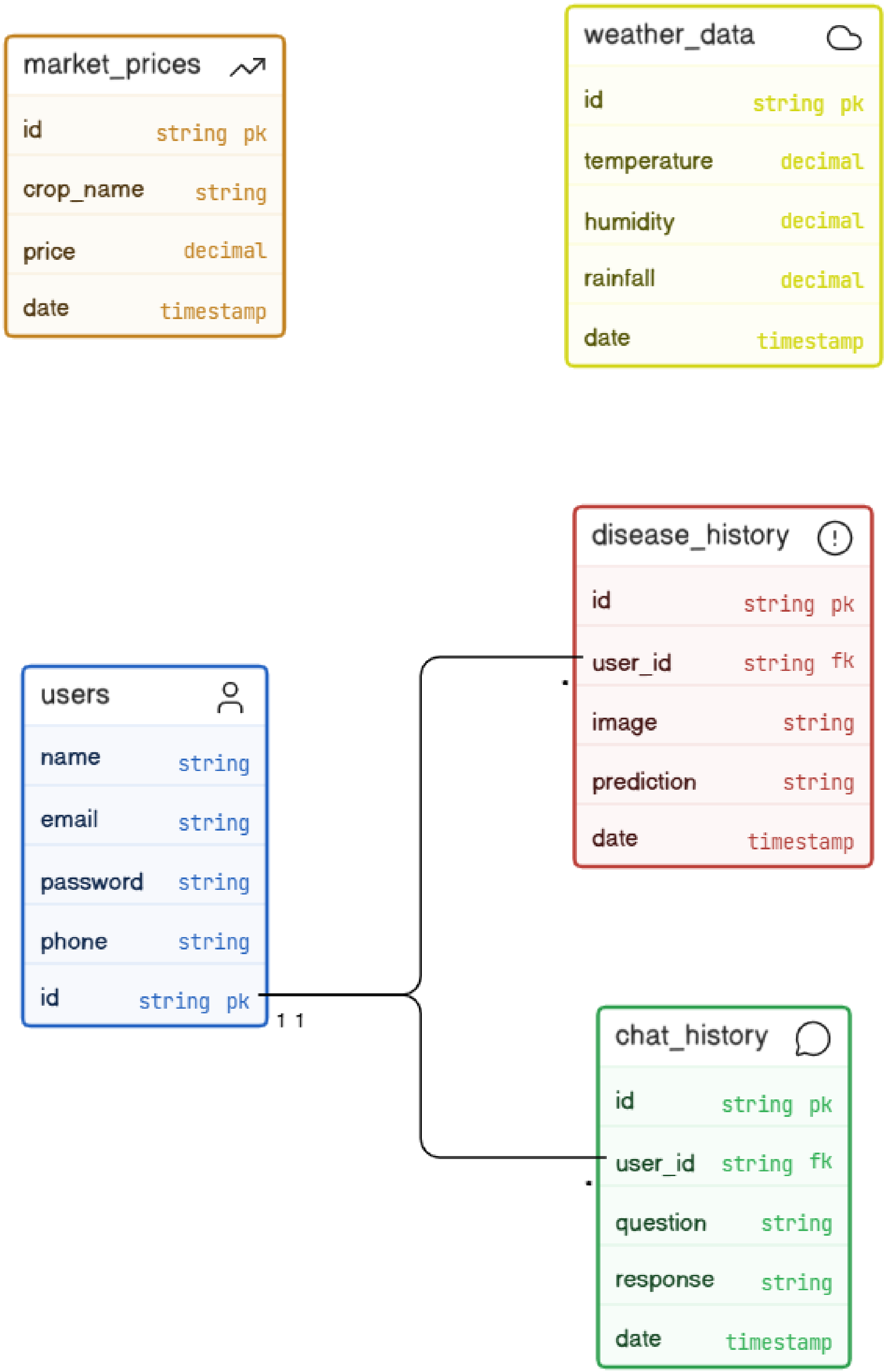

Fig-04: ER Diagram

v. Activity Diagram

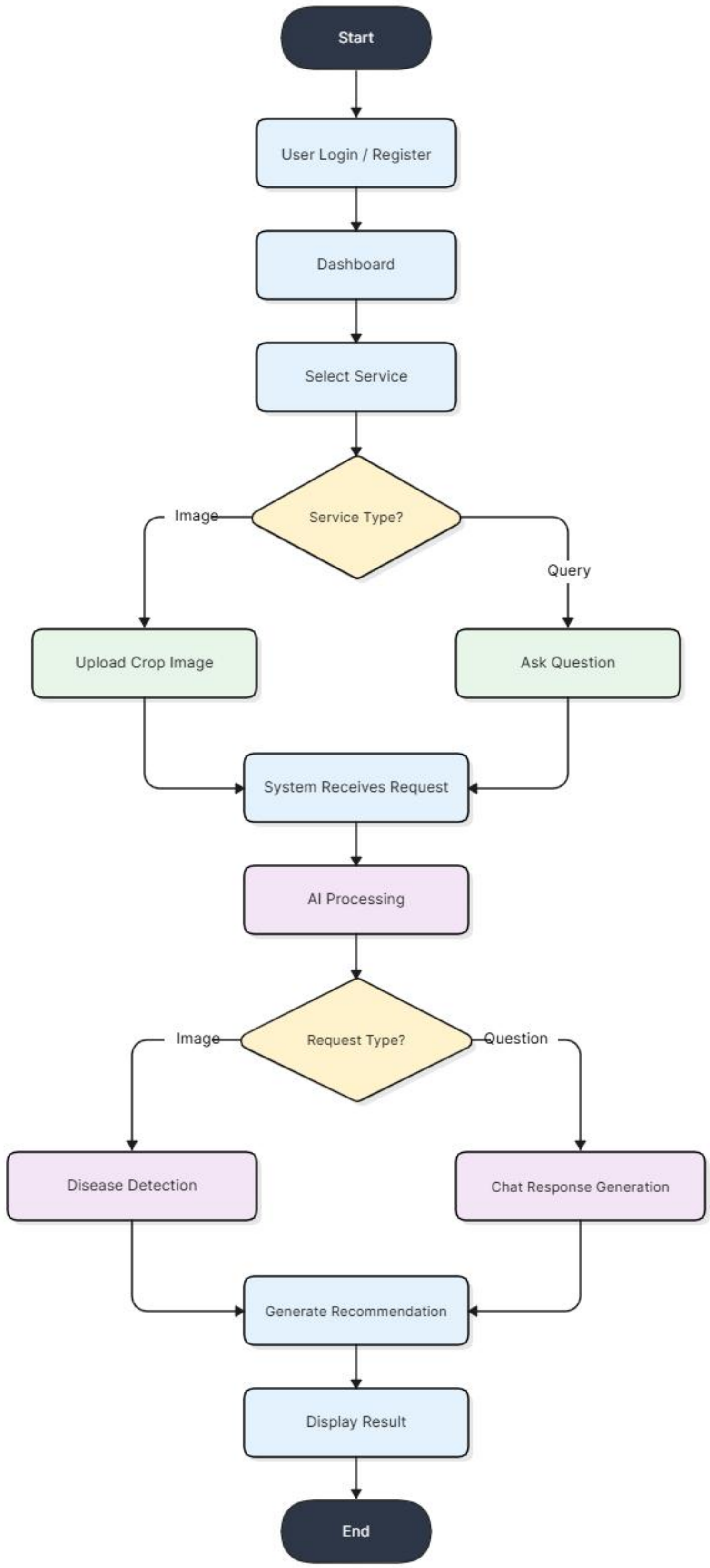

Fig-05: Activity Diagram

vi. User Interaction Diagram

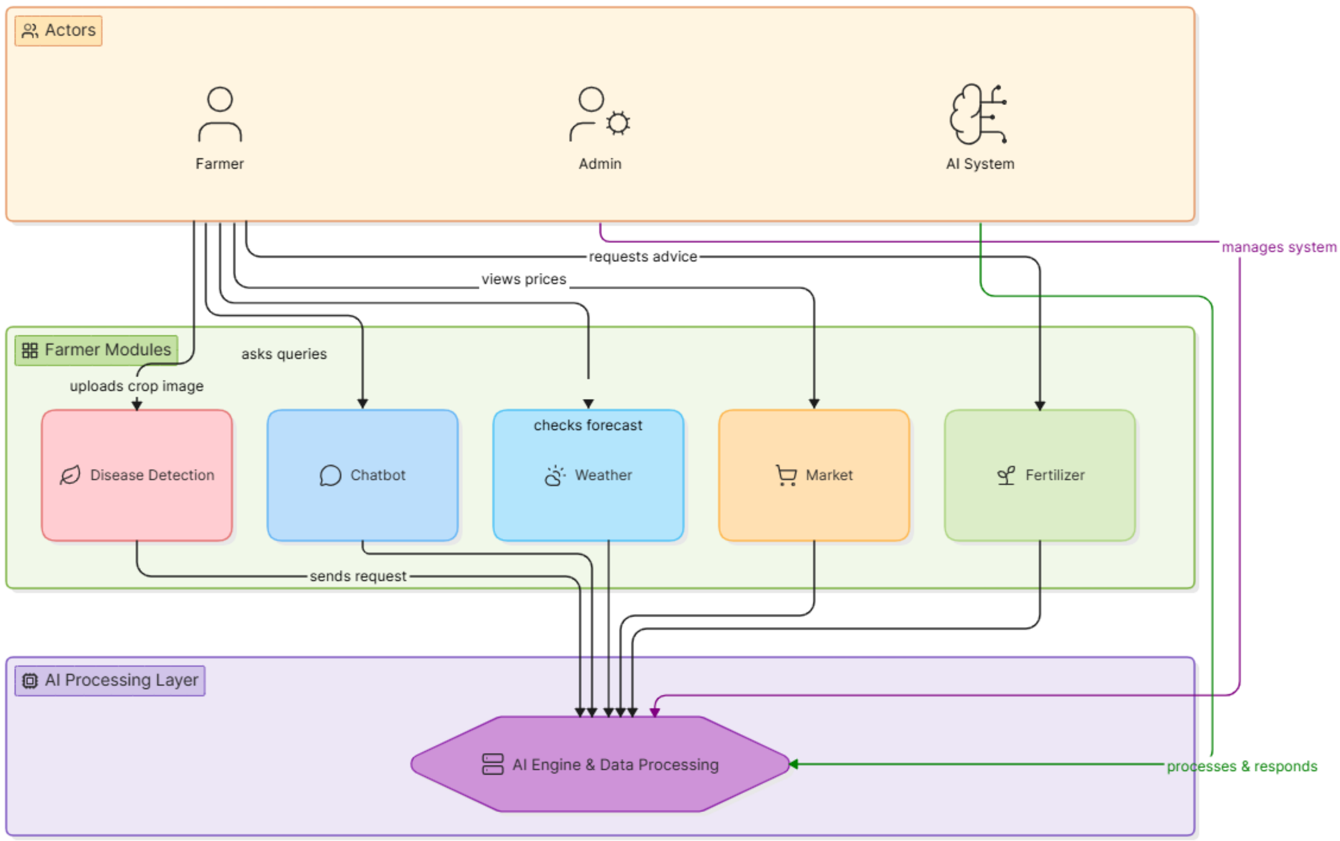

Fig-06: User Interaction Diagram

vii. Module Structure Diagram

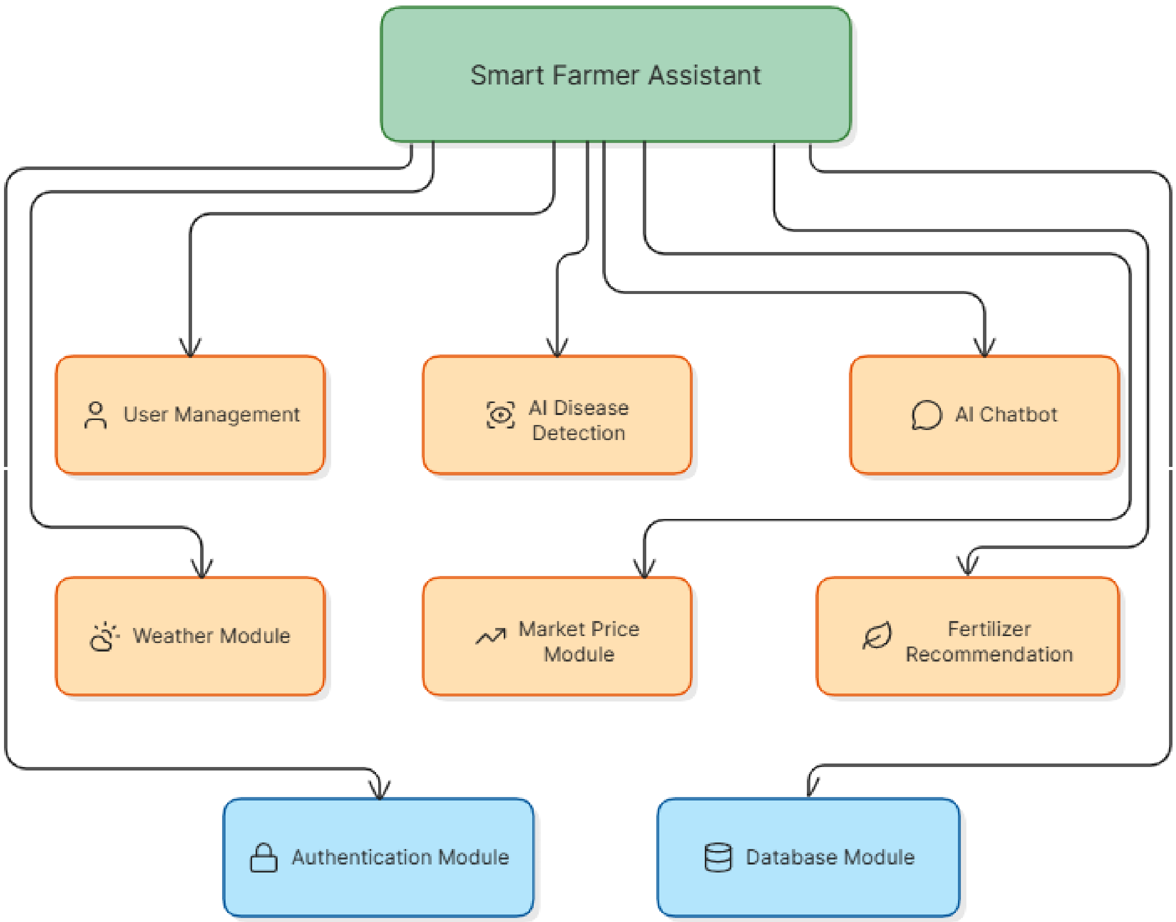

Fig-07: Module Structure Diagram